Test Report
Report **18264926**v3
Customer **Laboratorio Vidatec**
Customer Reference # DEMO

teletest
Lab Analisis

Sample Name # DEMOSTRACION GNT. SALIVA METABOLIZER DETOX
DEMO

Sample Number **18264926** Receipt Data: 30/06/2026 12:46

Report Validation and Issue Date **30/06/2026 12:46** Laboratorio

GNT. Detoxification



DEMOSTRACION GNT. SALIVA METABOLIZER DETOX

Date **30-06-2026**

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GNT. Detoxification

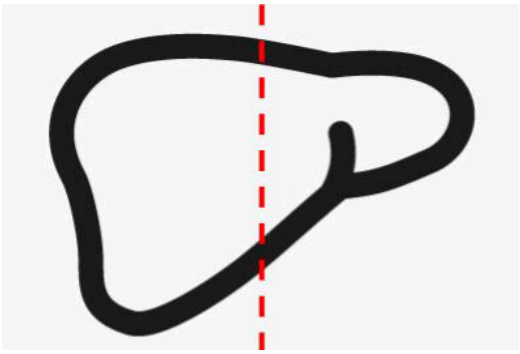
Interpretations are intended for healthcare professionals for informational purposes only and should not be interpreted as a diagnosis or treatment recommendation. Diagnosis and treatment decisions are the responsibility of the healthcare professional.
The accuracy of genetic tests is not 100%. Genetic results should be interpreted in the clinical context. The prevalence and significance of some allelic variations may be population-specific. Our reports are interpreted according to the European population.
Any positive test result indicates a genetic predisposition, but there may be additional risks not measured by the test.
Negative results do not imply the absence of genetic risk. DNA sequencing is used to detect polymorphisms; its sensitivity and specificity are 99%.

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Liver DETOXIFICATION. Phase I

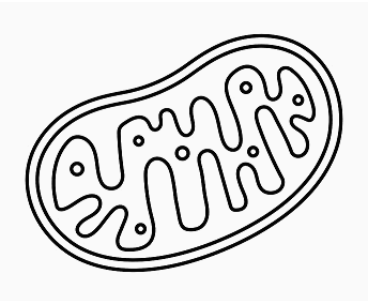
- Hydroxylation. Sterols
- Hydroxylation. Xenobiotics
- Hydroxylation. Fatty acids
- Hydroxylation. Eicosanoids



Liver DETOXIFICATION. Phase II

- Methylation
- Acetylation
- Glucurunization
- Sulfation

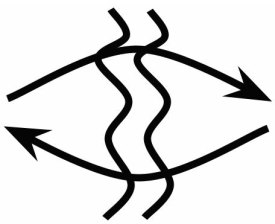
Oxidative stress



- Catalase
- Superoxide dismutase
- Glutathione
- DNA/RNA

Conveyors

- MTRNR1
- CFTR
- SLC15A2
- SLCO1B1
- SLCO2B1



Sample type Saliva

Cytochrome p450

Gene	Interpretation	Phenotype	Genotype/Haplotype
CYP1A1	Metabolizador normal	normal/normal	*1/*2C

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●	**CYP1A2**	Metabolizador normal	normal/normal	*1F/*1F
▼	**CYP1B1**	**Metabolizador intermedio**	normal/decreased	*1/*4
▼	**CYP2A6**	**Metabolizador intermedio**	normal/decreased	*1/*1H
▼	**CYP2B6**	**Metabolizador intermedio**	normal/decreased	*1/*6
●	**CYP2C8**	Metabolizador normal	normal/normal	*1A/*1A
▼	**CYP2C9**	**Metabolizador intermedio**	normal/decreased	*1/*3
●	**CYP2C19**	**Metabolizador intermedio**	no/increased	*2/*17
●	**CYP2D6**	Metabolizador normal	normal(1)/normal(1)	*1/*2
⚠	**CYP2D6enh**	**Promotor +**	normal/increased	AA/GG
*	**CYP2E1**	**Metabolizador intermedio**	normal/unknown	*7A/*7C+1B,*7A+1B/*7C
●	**CYP3A4**	Metabolizador normal	normal/normal	*1/*1
▼	**CYP3A5**	**Metabolizador intermedio**	normal/no	*1/*3
●	**CYP3A7**	Metabolizador normal	normal/normal	*1A/*1A
●	**CYP4B1**	Metabolizador normal	normal/normal	*1/*1
●	**CYP4F2**	Metabolizador normal	normal/normal	*1/*1
▼	**CYP5A1 (or TBXAS1)**	**Metabolizador intermedio**	normal/decreased	*1B/*8
●	**CYP8A1 (or PTGIS)**	Metabolizador normal	normal/normal	*1/*1
●	**CYP19A1**	Metabolizador normal	normal/normal	*1/*1
▼	**FMO1**	**Metabolizador intermedio**	normal/decreased	*1/*6
●	**POR**	Metabolizador normal	normal/normal	*1/*1

Methylation

	Gene	Interpretation	Phenotype	Genotype/Haplotype
▼	**COMT**	**Metabolizador pobre**	no/no	ValB/ValB

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MTHFR	**Poor metabolizer**			
- rs1801131		normal/disminuido	TG	
- rs1801133		muy disminuido/normal	AG	
MTR. rs1805087		normal/normal	AA	
MTRR. rs1801394	**Intermediate metabolizer**	disminuido/normal	AG	

Acetylation

Gene	Interpretation	Phenotype	Genotype/Haplotype
NAT2	**Metabolizador intermedio**	rapid/slow	*4/*6
NAT1	**Metabolizador ultrarrápido**	rapid/rapid	*4/*4

Glucuronidation

Gene	Interpretation	Phenotype	Genotype/Haplotype
UGT1A1	**Metabolizador intermedio**	normal/decreased	*1/*28+60+80+93
UGT1A3	**Metabolizador intermedio**	normal/decreased	*1a/*2a
UGT1A4	Metabolizador normal	normal/normal	*1/*1
UGT1A6	**Metabolizador rápido**	normal/increased	*1/*2
UGT1A7	**Metabolizador intermedio**	normal/decreased	*1/*4
UGT1A8	**Metabolizador rápido**	normal/increased	*1a/*2
UGT1A9	Metabolizador normal	normal/normal	*1hap_I/*1hap_III
UGT1A10	Metabolizador normal	normal/normal	*1/*1
UGT2B7	**Metabolizador rápido**	normal/increased	*1a/*2a
UGT2B15	**Metabolizador intermedio**	normal/decreased	*1/*2
UGT2B17	**Metabolizador intermedio**	normal/no	*1/*2

Sulfation

Gene	Interpretation	Phenotype	Genotype/Haplotype
SULT1A1	**Metabolizador intermedio**	normal/decreased	*1/*2

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SULT1A2	Metabolizador normal	normal/normal	WT/WT
SULT1A4	Metabolizador normal	normal/normal	WT/WT
SULT1E1. rs3736599	Metabolizador normal	normal/normal	CC

Glutathione

Gene	Interpretation	Phenotype	Genotype/Haplotype
GSTM1	**Metabolizador intermedio**	no/normal	Null/Present
GSTM2	Metabolizador normal	normal/normal	WT/WT
GSTM3	Metabolizador normal	normal/normal	CC
GSTM4	Metabolizador normal	normal/normal	TT
GSTP1	**Metabolizador intermedio**	normal/decreased	*A/*B
GSTT1	**Metabolizador intermedio**	normal/null	*A/Null
GSTA1. rs3957357	Metabolizador normal	normal/normal	GG

Oxidative stress

Gene	Interpretation	Phenotype	Genotype/Haplotype
CAT	Metabolizador normal	normal/normal	AA
SOD1	Metabolizador normal	normal/normal	CC
SOD2	**Metabolizador intermedio**	normal/disminuido	AG
SOD3	Metabolizador normal	normal/normal	CC
OGG1	**Metabolizador intermedio**	normal/decreased	WT/VAR
CDA	**Metabolizador rápido**	normal/increased	*1/*2
DPYD	Metabolizador normal	normal/normal	c.=/c.=
NQO1. rs1800566	**Intermediate metabolizer**	disminuido/normal	AG

Conveyors

Gene	Interpretation	Phenotype	Genotype/Haplotype

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CFTR	Transporte normal	normal/normal	WT/WT
MTRNR1	Transporte normal	null/normal	Null/m.=
SLC15A2	**Transporte alterado**	normal/high	*1/*2
SLCO1B1	Transporte normal	normal/normal	*1A/*1A
SLCO2B1	Transporte normal	normal/normal	*1/*1
ABCB1	**Altered transport**		
- rs1045642		disminuido/normal	AG
- rs1128503		disminuido/normal	AG
- rs2032582		disminuido/normal	AC
ABCC2	**Altered transport**		
- rs2273697		disminuido/normal	AG
- rs717620		normal/normal	CC

GNT. Detoxification

*(**RV:** Valeurs de référence exprimées en cursive)*

Access to digital report.
teletest.es->Mis Resultados

Report Validated by **Dra Concepción Alomar**

Report Validation and Issue Date 30-06-2026 12:46

Report issued by: **TeleTest Analytika S.A.**. Tel 932 123 345; laboratory@teletest.es

18264926 **43YMe**

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Clinical Analysis

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Reference values. Summary

GNT. Detoxification

Cytochrome p450

CYP1A1

The CYP1A1 gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase located in the endoplasmic reticulum, and its expression is induced by some polycyclic aromatic hydrocarbons. The enzyme's endogenous substrate is unknown; however, it is capable of metabolizing some PAHs into carcinogenic intermediates.

CYP1A2

The CYP1A2 gene encodes a member of the cytochrome P450 enzyme superfamily. It is a monooxygenase located in the endoplasmic reticulum, and its expression is induced by some polycyclic aromatic hydrocarbons (PAHs). The endogenous substrate of the enzyme is unknown; however, it is capable of metabolizing some PAHs into carcinogenic intermediates. Other xenobiotic substrates for this enzyme include caffeine, aflatoxin B1, and paracetamol.

CYP1B1

The CYP1B1 gene encodes a member of the cytochrome P450 enzyme superfamily. It is a monooxygenase located in the endoplasmic reticulum and metabolizes procarcinogens such as polycyclic aromatic hydrocarbons and 17beta-estradiol. Mutations in this gene are associated with primary congenital glaucoma.

CYP2A6

The CYP2A6 gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase located in the endoplasmic reticulum, and its expression is induced by phenobarbital. It also hydroxylates coumarin and metabolizes nicotine, aflatoxin B1, nitrosamines, and some pharmaceuticals.

CYP2B6

The CYP2B6 gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase located in the endoplasmic reticulum, and its expression is induced by phenobarbital. It is known to metabolize some xenobiotics and anticancer drugs such as cyclophosphamide and ifosfamide.

CYP2C8

The CYP2C8 gene encodes a member of the cytochrome p450 enzyme superfamily. It is a monooxygenase found in the endoplasmic reticulum, and its expression is induced by phenobarbital. It metabolizes many xenobiotics, including mephenytoin, benzopyrene, 7-ethioxycoumarin, and taxol.

CYP2C9

The CYP2C9 gene encodes a member of the cytochrome p450 enzyme superfamily. It is a monooxygenase found in the endoplasmic reticulum, and its expression is induced by rifampicin. It metabolizes many xenobiotics, including phenytoin, tolbutamide, ibuprofen, and S-warfarin.

CYP2C19

The CYP2C19 gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase located in the endoplasmic reticulum. It metabolizes xenobiotics such as mephenytoin, omeprazole, diazepam, and some barbiturates.

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CYP2D6

The CYP2D6 gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase located in the endoplasmic reticulum. It metabolizes up to 25% of common drugs such as antidepressants, antipsychotics, analgesics, antitussives, beta-adrenergic blockers, antiarrhythmics, and antiemetics.

CYP2D6enh

The region represents an enhancer of the CYP2D6 gene. Mutations cause increased promoter activity, making it hypersensitive to DNAse I.

CYP2E1

The CYP2E1 gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase located in the endoplasmic reticulum, and its expression is induced by ethanol, diabetes, and starvation. It metabolizes endogenous substrates (mainly ethanol, acetone, and acetal) and exogenous substrates such as benzene, carbon tetrachloride, ethylene glycol, and nitrosamines (which are premutagens found in cigarette smoke).

CYP3A4

The CYP3A4 gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase located in the endoplasmic reticulum, and its expression is induced by glucocorticoids and some pharmacological agents. It metabolizes paracetamol, codeine, cyclosporine A, diazepam, erythromycin, and chloroquine. It also processes some steroids and carcinogens.

CYP3A5

The CYP3A5 gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase that metabolizes both drugs and steroid hormones (primarily testosterone and progesterone). Expression of this gene is highly variable, and polymorphisms are associated with susceptibility to hypertension.

CYP3A7

The CYP3A7 gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase located in the endoplasmic reticulum. It hydroxylates testosterone and dehydroepiandrosterone 3-sulfate, which is involved in estriol formation during pregnancy.

CYP4B1

The CYP4B1 gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase located in the endoplasmic reticulum and metabolizes certain carcinogens.

CYP4F2

The CYP4F2 gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase located in the endoplasmic reticulum. The enzyme initiates the inactivation and degradation of leukotriene B4, a potent mediator of inflammation.

CYP5A1 (or TBXAS1)

The CYP5A1 or TBXAS1 gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase located in the endoplasmic reticulum. It catalyzes the conversion of prostaglandin H2 to thromboxane A2, a potent vasoconstrictor and inducer of platelet aggregation. The enzyme plays a role in several pathophysiological processes, including hemostasis, cardiovascular disease, and stroke.

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CYP8A1 (or PTGIS)

The CYP8A1 or PTGIS gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase located in the endoplasmic reticulum and catalyzes the conversion of prostaglandin H2 to prostacyclin (prostaglandin I2), a potent vasodilator and inhibitor of platelet aggregation. An imbalance between prostacyclin and its physiological antagonist, thromboxane A2, contributes to the development of myocardial infarction, stroke, and atherosclerosis.

CYP19A1

The CYP19A1 gene encodes a member of the cytochrome P450 enzyme superfamily. This protein is a monooxygenase located in the endoplasmic reticulum and catalyzes the final steps of estrogen biosynthesis. Mutations in this gene can result in either increased or decreased aromatase activity.

FMO1

The FMO1 gene encodes a member of the flavin-containing monooxygenase family. This enzyme type is NADPH-dependent and catalyzes the oxidation of soft nucleophilic heteroatom centers in pharmaceuticals, pesticides, and xenobiotics.

POR

The POR gene encodes an endoplasmic reticulum membrane oxidoreductase that is essential for multiple metabolic processes, including reactions catalyzed by cytochrome P450 proteins. The encoded protein has a flavin adenine dinucleotide (FAD)-binding domain and a flavodoxin-like domain that bind two cofactors, FAD and FMN, allowing it to donate electrons directly from NADPH to all microsomal P450 enzymes. Mutations in this gene cause different pathologies such as amenorrhea, disordered steroidogenesis, congenital adrenal hyperplasia, and Antley-Bixler syndrome.

Methylation

COMT

The COMT (catechol-O-methyltransferase) gene encodes an enzyme that catalyzes the transfer of a methyl group from S-adenosylmethionine to catecholamines (dopamine, adrenaline, and noradrenaline). In addition to its role in the metabolism of endogenous substances, COMT is important in the metabolism of catechol drugs used in the treatment of hypertension, asthma, and Parkinson's disease. COMT is found in two forms in tissues: a soluble form (S-COMT) and a membrane-bound form (MB-COMT).

MTHFR

The MTHFR gene encodes a protein that catalyzes the conversion of 5,10-methylenetetrahydrofolate to 5-methyltetrahydrofolate, a cosubstrate for the remethylation of homocysteine to methionine. Genetic variation in this gene influences susceptibility to vascular occlusive disease, neural tube defects, colon cancer, and acute leukemia, and mutations in this gene are associated with methylenetetrahydrofolate reductase deficiency.

Acetylation

NAT2

The NAT2 gene encodes N-acetyltransferase 2. This protein catalyzes the transfer of an acetyl group from acetyl-CoA to various arylamine and hydrazine substrates. It is involved in the metabolism of drugs and xenobiotics.

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NAT1

The NAT1 gene encodes N-acetyltransferase 1. This protein catalyzes the transfer of an acetyl group from acetyl-CoA to various arylamine and hydrazine substrates. It is involved in the metabolism of drugs and xenobiotics. It also regulates folic acid catabolism.

Glucuronidation

UGT1A1

The UGT1A1 gene encodes a UDP-glucuronosyltransferase, an enzyme of the glucuronidation pathway that transforms small lipophilic molecules, such as steroids, bilirubin, hormones, and drugs, into water-soluble excretable metabolites. This gene is part of a complex locus encoding several UDP-glucuronosyltransferases. The preferred substrate of this enzyme is bilirubin, although it also has moderate activity with simple phenols, flavones, and C18 steroids. Mutations in this gene result in Crigler-Najjar syndrome types I and II and Gilbert syndrome.

UGT1A3

The UGT1A3 gene encodes a UDP-glucuronosyltransferase, an enzyme of the glucuronidation pathway that transforms small lipophilic molecules, such as steroids, bilirubin, hormones, and drugs, into water-soluble excretable metabolites. This gene is part of a complex locus encoding several UDP-glucuronosyltransferases. The preferred substrate of this enzyme is estrone, 2-hydroxyestrone, and benzo alpha-pyrene metabolites.

UGT1A4

The UGT1A4 gene encodes a UDP-glucuronosyltransferase, an enzyme of the glucuronidation pathway that transforms small lipophilic molecules, such as steroids, bilirubin, hormones, and drugs, into water-soluble excretable metabolites. This gene is part of a complex locus encoding several UDP-glucuronosyltransferases. The preferred substrates of this enzyme are amines, steroids, and sapogenins, although it can also metabolize bilirubin.

UGT1A6

The UGT1A6 gene encodes a UDP-glucuronosyltransferase, an enzyme in the glucuronidation pathway that converts small lipophilic molecules, such as steroids, bilirubin, hormones, and drugs, into water-soluble excretable metabolites. This gene is part of a complex locus encoding several UDP-glucuronosyltransferases. This enzyme is activated by phenolic and planar compounds.

UGT1A7

The UGT1A7 gene encodes a UDP-glucuronosyltransferase, an enzyme in the glucuronidation pathway that transforms small lipophilic molecules, such as steroids, bilirubin, hormones, and drugs, into water-soluble excretable metabolites. This gene is part of a complex locus encoding several UDP-glucuronosyltransferases. Its activity is modulated by phenol levels.

UGT1A8

The UGT1A8 gene encodes a UDP-glucuronosyltransferase, an enzyme of the glucuronidation pathway that transforms small lipophilic molecules, such as steroids, bilirubin, hormones, and drugs, into water-soluble excretable metabolites. This gene is part of a complex locus encoding several UDP-glucuronosyltransferases. The preferred substrates of this enzyme are coumarins, phenols, anthraquinones, flavones, and some opioids.

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UGT1A9

The UGT1A9 gene encodes a UDP-glucuronosyltransferase, an enzyme in the glucuronidation pathway that transforms small lipophilic molecules, such as steroids, bilirubin, hormones, and drugs, into water-soluble excretable metabolites. This gene is part of a complex locus encoding several UDP-glucuronosyltransferases. The preferred substrate is phenols and/or phenolic derivatives.

UGT1A10

The UGT1A10 gene encodes a UDP-glucuronosyltransferase, an enzyme of the glucuronidation pathway that transforms small lipophilic molecules, such as steroids, bilirubin, hormones, and drugs, into water-soluble excretable metabolites. This gene is part of a complex locus encoding several UDP-glucuronosyltransferases. The preferred substrate of this enzyme is mycophenolic acid, coumarins, and quinolines.

UGT2B7

The UGT2B7 gene encodes a protein of the UDP-glycosyltransferase (UGT) family. This protein is located in the microsomal membrane and has a unique specificity for the estrogens 3,4-catechol and estriol, suggesting that it may play an important role in regulating the level and activity of these potent estrogen metabolites.

UGT2B15

The UGT2B15 gene encodes a glycosyltransferase involved in the metabolism and elimination of toxic compounds, both endogenous and xenobiotic. This gene plays a role in the regulation of estrogens and androgens.

UGT2B17

The UGT2B17 gene encodes a UDP-glucuronosyltransferase, an enzyme in the glucuronidation pathway. Copy number variations in this gene are associated with increased susceptibility to osteoporosis.

Sulfation

SULT1A1

The SULT1A1 gene encodes a sulfotransferase, which catalyzes the sulfate conjugation of many hormones, neurotransmitters, drugs, and xenobiotic compounds. These cytosolic enzymes differ in their tissue distributions and substrate specificities. This gene encodes one of two phenol sulfotransferases with thermostable enzymatic activity.

SULT1A2

The SULT1A2 gene encodes a sulfotransferase, which catalyzes the sulfate conjugation of many hormones, neurotransmitters, drugs, and xenobiotic compounds. These cytosolic enzymes differ in their tissue distributions and substrate specificities. This gene encodes one of two phenol sulfotransferases with thermostable enzymatic activity.

SULT1A4

The SULT1A4 gene encodes a sulfotransferase, which catalyzes the sulfate conjugation of many hormones, neurotransmitters, drugs, and xenobiotic compounds. These cytosolic enzymes differ in their tissue distributions and substrate specificities. This gene encodes a phenol sulfotransferase with thermolabile enzymatic activity.

SULT1E1. rs3736599

The SULT1E1 gene encodes estrogen sulfotransferase, a cytoplasmic enzyme that catalyzes the sulfoconjugation of active estrogens (estradiol, estrone) to their inactive, water-soluble sulfated forms, facilitating their elimination. SULT1E1 plays a key role in regulating active estrogen levels at the tissue level and in inactivating potentially genotoxic estrogenic metabolites, such as catechol-estrogens.

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Glutathione

GSTM1

The GSTM1 gene encodes the glutathione-S-transferase mu protein. This protein is located in both the cytosol and the membrane. The mu class is involved in the detoxification of electrolyte compounds such as carcinogens, environmental toxins, and the byproducts of oxidative stress.

GSTM2

The GSTM2 gene encodes the glutathione-S-transferase mu protein. This protein is located in both the cytosol and the membrane. The mu class is involved in the detoxification of electrolyte compounds such as carcinogens, environmental toxins, and the byproducts of oxidative stress.

GSTM3

The GSTM3 gene encodes the glutathione-S-transferase mu protein. This protein is located in both the cytosol and the membrane. The mu class is involved in the detoxification of electrolyte compounds such as carcinogens, environmental toxins, and the byproducts of oxidative stress.

GSTM4

The GSTM4 gene encodes the glutathione-S-transferase mu protein. This protein is located in both the cytosol and the membrane. The mu class is involved in the detoxification of electrolyte compounds such as carcinogens, environmental toxins, and the byproducts of oxidative stress.

GSTP1

The GSTP1 gene encodes a glutathione-S-transferase of the Pi family. This protein detoxifies many hydrophobic and electrophilic compounds with glutathione.

GSTT1

The GSTT1 gene encodes glutathione S-transferase theta 1, which belongs to a superfamily of proteins that catalyze the conjugation of reduced glutathione to a variety of electrophilic and hydrophobic compounds. The GSTT1 gene is specific and is absent in 38% of the population.

GSTA1. rs3957357

 The GSTA1 gene encodes glutathione S-transferase alpha 1, the most abundant hepatic isoform of the GST family. This enzyme catalyzes the conjugation of reduced glutathione with electrophilic compounds, including lipid peroxidation products, reactive metabolites generated by Phase I enzymes, carcinogens such as aflatoxin B1, and various environmental toxins. GSTA1 expression is regulated by polymorphisms in the gene's promoter region.

Oxidative stress

CAT

The CAT gene encodes a heme antioxidant enzyme present in the peroxisome of almost all aerobic cells. Catalase converts hydrogen peroxide from the reactive oxygen species (ROS) into water and oxygen. Polymorphisms in this gene have been associated with decreased catalase activity. Acatalasemia is the only known disease caused by this gene.

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SOD1

The SOD1 gene encodes superoxide dismutase. The protein binds copper and zinc ions. It is composed of two isoenzymes, which are responsible for destroying superoxide free radicals in the body. One isoenzyme is located in the cytoplasm and the other in the mitochondrial membrane. Both metabolize oxygen free radicals into hydrogen peroxide. In addition, this protein contains an antimicrobial peptide that displays antibacterial, antifungal, and anti-MRSA activity against E. coli, E. faecalis, S. aureus, S. aureus MRSA LPV+, S. agalactiae, and the yeast C. krusei. Mutations in this gene have been implicated as causes of familial amyotrophic lateral sclerosis.

SOD2

The SOD2 gene encodes superoxide dismutase. The protein binds iron and manganese ions. It encodes a mitochondrial protein that forms a homotetramer and binds one manganese ion per subunit. This protein binds superoxide byproducts of oxidative phosphorylation and converts them into hydrogen peroxide and diatomic oxygen. Mutations in this gene have been associated with idiopathic cardiomyopathy (IDC), premature aging, sporadic motor neuron disease, and cancer.

SOD3

The SOD3 gene encodes superoxide dismutase. Proteolytic processing of the encoded protein results in the formation of two distinct homotetramers that differ in their ability to interact with the extracellular matrix (ECM). Homotetramers consisting of the intact protein, or C-type subunit, exhibit high affinity for heparin and are anchored to the ECM. Homotetramers consisting of a proteolytically cleaved form of the protein, or A-type subunit, exhibit low affinity for heparin and do not interact with the ECM. A mutation in this gene may be associated with an increased risk of heart disease.

OGG1

The OGG1 gene encodes an enzyme that cleaves 8-oxoguanine, a mutagenic base byproduct produced by exposure to reactive oxygen. This enzyme's action includes chain-scission lyase activity.

CDA

The CDA gene encodes an enzyme involved in pyrimidine salvage. The encoded protein forms a homotetramer that catalyzes the irreversible hydrolytic deamination of cytidine and deoxycytidine to uridine and deoxyuridine, respectively. Mutations are associated with decreased sensitivity to the cytosine arabinoside, a cytosine nucleoside analogue.

DPYD

The DPYD gene encodes an enzyme that catabolizes pyrimidine, uracil, and thymidine. Mutations in this gene result in dihydropyrimidine dehydrogenase deficiency, an error in pyrimidine metabolism associated with thymine-uraciluria, and an increased risk of toxicity in cancer patients receiving 5-fluorouracil chemotherapy.

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NQO1. rs1800566

The NQO1 gene belongs to the NAD(P)H (quinone) dehydrogenase family and encodes a 2-electron cytoplasmic reductase. This FAD-binding protein forms homodimers and reduces quinones to hydroquinones. The enzymatic activity of this protein prevents the one-electron reduction of quinones that leads to the production of radical species. Mutations in this gene have been associated with tardive dyskinesia, an increased risk of hematotoxicity following exposure to benzene, and susceptibility to several forms of cancer. Altered expression of this protein has been observed in many tumors and is also associated with Alzheimer's disease.

[provided by RefSeq, Jul 2008]

Conveyors

CFTR

The CFTR gene encodes a member of the ATP transporter (ABC) family. The protein functions as a chloride channel and controls the secretion and absorption of ions and water in epithelial tissues. Channel activation is mediated by cycles of phosphorylation of the regulatory domain, ATP binding by the nucleotide-binding domains, and ATP hydrolysis. Mutations in this gene cause cystic fibrosis. Multiple pseudogenes have been identified in the human genome.

MTRNR1

The MTRNR1 gene encodes a protein that binds to DNA and DNA transcription factors. It is responsible for regulating osteoblast proliferation, carbohydrate utilization, and phosphate-consuming metabolic processes. It is located in both the mitochondria and the nucleus.

SLC15A2

The SLC15A2 gene encodes a proton-coupled peptide transporter responsible for the absorption of small peptides, as well as beta-lactam antibiotics and other peptide-like drugs from the tubular filtrate. It is primarily found in the small intestine.

SLCO1B1

The SLCO1B1 gene encodes a specific member of the organic anion transporter family. This protein is a transmembrane receptor that mediates the sodium-independent uptake of numerous endogenous compounds, such as bilirubin, 17-beta-glucuronosyl estradiol, and leukotriene C4. It is also involved in the elimination of pharmacological compounds such as statins, bromosulfophthalein, and rifampin from the blood into hepatocytes.

SLCO2B1

The SLCO2B1 gene encodes a specific member of the organic anion transporter family. This protein is involved in regulating the placental uptake of sulfated steroids.

ABCB1

The ABCB1 gene encodes P-glycoprotein (P-gp), also known as multidrug resistance protein 1 (MDR1). It is an ATP-dependent efflux transporter located in the plasma membrane of intestinal epithelial cells, hepatocytes, the proximal renal tubule, the blood-brain barrier, and the placenta. Its main function is to limit the absorption of xenobiotics from the intestinal lumen, facilitate the biliary and renal excretion of toxic compounds, and protect sensitive tissues (brain, fetus, testes) from exposure to harmful substances.

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ABCC2

The ABCC2 gene encodes for multidrug resistance protein 2 (MRP2), an ATP-dependent efflux transporter located in the apical membrane of hepatocytes. Its main function is the active export of glutathione, glucuronide, and sulfate conjugates —that is, the end products of Phase II reactions—from the hepatocyte into the bile for elimination.

Legend Symbols



Low .... Reference Value .... High

Interpret according to patient data

Result with comments

Indeterminate or Inconclusive

Result
VR Max
VR Min

Antecedent, if applicable



Reference Value

Result

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